

Assignment 1: Lists and Operators

1. Define the relation `flatten( List, Flatlist)` where `List` can be a list of lists, and `Flatlist` is `List` ,flattened' so that the elements of `List`'s sublists (or sub-sublists) are reorganized as one plain list.

For example: `?- flatten([a,b,[c,d],[[]],[[e]]],f), L).`

`L = [a,b,c,d,e,f]`

2. Define the relation `translate(List1, List2)` to translate a list of numbers between 0 and 9 to a list of the corresponding words.

For example: `translate([3, 5, 1, 3], [three,five,one,three])`

Use the following as an auxiliary relation:

`means( 0, zero). means(1, one). means(2, two). ...`

3. Suggest an appropriate definition of operators (`was, of, the`) to be able to write clauses like `, joe was the head of the department.`

and then ask Prolog: `?- Who was the head of the department.`

`Who = joe`

`?- joe was What.`

`What = the head of the department`

4. Define the predicate `subsum(NumList, Sum, PartList)` so that `NumList` is a list of numbers, `PartList` is a subset of these numbers, and the sum of the numbers in `PartList` is `Sum`.

For example: `?- subsum([1,2,7,3,6],6, PartList).`

`PartList = [1,2,3];`

`PartList = [6];`

`...`

5. Define the procedure `between(Num1, Num2, X)` which, for two given integers `Num1` and `Num2`, generates through backtracking all the integers `X` that satisfy the constraint `Num1 <= X <= Num2`.